The True Cost of Providing Liquidity in Uniswap v3

Decomposing fee income against loss-versus-rebalancing across seven pools and 43 months, and testing whether a closed-form price for adverse selection becomes a tradable edge

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ABSTRACT

A liquidity provider on a constant-function market maker has no oracle and quotes continuously, so informed traders arbitrage the pool against the wider market and the provider pays for it. Milionis, Moallemi, Roughgarden and Zhang [1] price that cost in closed form as loss-versus-rebalancing (LVR), the gap between the pool and a continuously rebalanced portfolio holding the same delta, which for a constant-product pool is $\sigma^2/8$ per unit value per unit time. Whether fee income covers it is an empirical question that is usually answered with pool-level aggregates, which conflate providers with different ranges and entry dates. We reconstruct a single \$100k position at ±10% from raw swap events on an Ethereum archive node — no vendor API, no subgraph — for seven Uniswap v3 pools spanning a 23× range of volatility, and decompose its return into fees, LVR and gas over 272 monthly cohorts. Fee accrual is validated against the pool's own *feeGrowthInside* accumulator read on-chain, and the reconstructed price against pool state. We find that liquidity provision is **not** uniformly loss-making: the stablecoin pool earns +8.43% a year with a confidence interval excluding zero, against -9.10% on the deepest ETH/USDC tier. The closed form explains around 62% of daily variation in realised LVR with zero fitted parameters, and its bias is monotone in volatility — realised over predicted runs 1.98 at a peg to 0.76 in the top percentile of days. But the economic reading usually placed on this does not survive: the negative cross-sectional association between volatility and net return is a between-pool composition effect that reverses sign within pools, and fee income scales with variance almost as steeply as LVR does, so volatility does not rank pools. No allocation rule we can construct clears a deflated-Sharpe bar, and decomposing the gap to a perfect-foresight ceiling shows why: perfect variance forecasting is worth 0.9 percentage points and perfect fee forecasting 11.4. **The cost side is priced; the uncertainty is all on the income side.**

Keywords: automated market makers, loss-versus-rebalancing, adverse selection, Uniswap v3, maximal extractable value, market microstructure.

1. Introduction

A market maker on a traditional venue quotes a spread, observes order flow, and withdraws when it turns toxic. A liquidity provider on a constant-function market maker can do none of these things. The pool quotes a price determined entirely by its own reserves, updates only when someone trades against it, and cannot decline a trade. When the external price moves, the pool is stale by construction, and the first arbitrageur to reach it takes the difference. The provider is paid a fee on that trade and loses the mispricing.

This is not impermanent loss. Impermanent loss compares the pool against holding the two assets, and so mixes the cost of providing liquidity with a directional bet on the pair. Milionis, Moallemi, Roughgarden and Zhang [1] isolate the former by comparing the pool against a portfolio that continuously rebalances to the same delta. The difference — loss-versus-rebalancing — is the cost of quoting without an oracle, it is always non-negative, and for a constant-product pool it has a closed form in the instantaneous variance alone. The practical consequence is that a provider's economics reduce to a single comparison: does fee income exceed LVR?

That question is usually answered with pool-level aggregates — total fees against total volume, or an advertised annual percentage rate. Those numbers describe the pool, not a provider. A Uniswap v3 pool contains providers with different ranges, different entry dates and different rebalancing policies, whose returns differ by more than the pool-level figure varies across pools. To say anything about whether providing liquidity pays, one has to fix a position and follow it.

This paper does that. We specify one position — \$100,000 deployed at ±10% around spot, re-centred when it exits — reconstruct it from raw swap events for seven pools over up to 43 months, and decompose its return into fee income, LVR and gas. The specification is deliberately unremarkable: it is what a competent provider would run, not an optimised strategy, and every parameter is varied in Section 6.7.

Three contributions follow. First, a measurement stack built entirely from primary sources — an Ethereum archive node and public exchange data, with no vendor API and no subgraph — in which every layer is validated against an independent one before anything is built on it. Second, a test of the closed form across a 23× range of volatility, which locates not just whether it holds but *where* it stops holding, in both directions. Third, and least comfortably, a set of retractions: the economic story most often told about this cost — that volatility determines whether liquidity provision pays — is a composition effect in our data, and we report the tests that dismantled it.

2. Why the cost is hard to observe

LVR is not recorded anywhere. It is a counterfactual: the difference between what a position earned and what a continuously rebalanced portfolio holding the same delta would have earned. Constructing it requires the pool's price path at block resolution, an external reference price on a compatible clock, and a position whose delta is known at every instant. Each is a source of silent error.

Uniswap v3 [2] holds the pool price on-chain as a Q64.96 fixed-point square root. Reading it as a floating-point number loses precision at exactly the magnitudes where the tick arithmetic is decided, so the entire price path here is carried in exact integers and decimals and never in binary floating point. The reference price must be aligned to block timestamps without using information dated after the block, which rules out the interpolation a naive join would perform. And the position's fee accrual depends on the pool's own accumulator semantics, which are defined in terms of growth *outside* the position's range and are not recoverable from swap volume alone.

The consequence is that a plausible-looking LVR series can be wrong by a factor with no symptom. Our response is to compute each quantity at least twice by independent routes and gate on their agreement, and to report the three faults this discipline caught in Section 7 — all of which produced valid-looking numbers.

3. Data

The universe is seven Uniswap v3 mainnet pools chosen to span volatility rather than liquidity: a stablecoin pair, gold against the dollar, a cross of two correlated majors, a major against the dollar at three fee tiers, and a second major against the dollar. Median annualised volatility ranges from 2.3% to 53.1% — a 23× spread, and 528× in variance, which is the quantity LVR scales with. A theory that holds across that range is saying something; one verified on a single pair is not.

The gold pool was added last and for one reason. The other six cluster at 2.3% and then at 28–53%, leaving the whole region between unobserved — and that is exactly where the claims in Section 6.4 had never been tested. PAXG/USDC runs near 20%. It carries a **shorter sample**: the pool had no liquidity before April 2025, so it contributes 16 monthly cohorts against 43 for the others. It is reported as a supplementary point and excluded from anything requiring the common window. That the crypto volatility range 3–28% is populated almost entirely by liquid-staking tokens and tokenised commodities — assets with either no centralised reference price or a thin pool — is itself the reason the region was empty.

Construction step	Observations	Note
Swap events decoded	10,461,521	1 Jan 2023 – 31 Jul 2026, seven pools
Blocks traversed	7,249,328	archive node, no vendor API
Pool-days with LVR and σ	8,181	empirical and analytical both present
Monthly position cohorts	272	$100k at ±10%, re-centred on exit
Pool-weeks in the forecast panel	1,187	188 weeks, lagged features only
CEX reference series	1-second klines	per-pool symbol; inverted where the venue quotes the pair the other way
Raw data retained	10 GB	regenerable from the fetchers

Table 1. Sample construction. Every filter is a documented configuration parameter rather than a hard-coded choice, and the whole chain rebuilds from the raw tape under a single command.

No commercial data was used and no API key was required. Swap events, pool state, fee accumulators and block base fees are read directly from an Ethereum archive node; reference prices and perpetual funding come from public exchange archives. The total monetary cost of the dataset was zero, which matters less as an economy than as a guarantee that every input is inspectable at source.

One consequential exclusion. Bundle-level MEV attribution requires historical relay data that the public Flashbots endpoint no longer serves for our sample period. Sandwich and arbitrage classification, in the sense of Daian et al. [3], is done from on-chain traces alone, which undercounts. The undercount is one-sided and we treat MEV magnitude as the least robust quantity in the paper; the decomposition does not rest on it, because MEV attribution is neutral with respect to the fees-minus-LVR identity.

4. Methodology

4.1 The decomposition identity

Everything rests on one accounting identity. The return of an AMM position equals the return of the continuously rebalanced portfolio holding the same delta, plus fees, minus LVR:

$$R_{AMM} = R_{rebalanced} + F - LVR \tag{1}$$

The first term is a directional exposure that any provider can hedge and that has nothing to do with market making. The second and third are the market-making business. We therefore report F − LVR − gas as the object of interest and call it net market-making return. It is important that this quantity is **already delta-hedged by construction**: no hedge profit-and-loss is added on top, because the identity has removed the directional term analytically rather than empirically. The identity is asserted as a runtime check on every cohort.

4.2 Loss-versus-rebalancing

For a position holding n(S) units of the risky asset at external price S, the instantaneous LVR rate is

$$dLVR/dt = (\sigma^2/2) \cdot S^2 \cdot |dn/dS| \tag{2}$$

which for a concentrated Uniswap v3 position with liquidity L reduces to $\sigma^2 L\sqrt{P}/4$, and for a full-range constant-product pool to the familiar $\sigma^2 V/8$ of [1]. We compute LVR three independent ways: from the external price path assuming perfect arbitrage; from the realised pool price path, which measures what arbitrageurs actually extracted; and analytically from the closed form fed a realised-variance estimate. The first is a fee-tier-blind upper bound by construction, the second is the object we regress on, and the third is the theory under test. Agreement between the first two is a gate; disagreement between the second and third is the subject of Section 6.2.

4.3 Fee accrual

Fees are accrued swap by swap over the position's range, pro-rated across tick crossings when a swap traverses a boundary, and scaled by the position's share of active liquidity. Two conventions matter. The share is $L/(L_{active} + L)$, not L/L_{active}, because a deposited position dilutes the pool it joins; the undiluted form is used only when reconciling against the pool's accumulator, which is defined before the deposit. And a swap that crosses a boundary has its fee pro-rated along the traversed tick path rather than assigned wholly inside or outside, which is where a single crossing swap can otherwise misstate a month.

4.4 Volatility

The closed form takes a variance, so a forecast of it is required wherever LVR is predicted rather than measured. We estimate realised variance from 5-minute sampling and compare six forecasts — persistence, RiskMetrics EWMA, the HAR family of Corsi [9], HAR-Q, HAR-CJ and HAR-RV augmented with Deribit's implied-volatility index — scored on QLIKE as well as root mean squared error, since variance is a latent right-skewed target on which squared error over-weights a handful of explosive days. HAR-Q [4] wins on both, lifting out-of-sample R^2 from 0.114 to 0.157, and is used throughout.

All forecasting uses purged and embargoed cross-validation [5], so no fold sees data adjacent to its own test window, and every specification executed during the study is logged automatically so the trial count cannot be reconstructed favourably after the fact.

5. Validation

Because the position is simulated rather than observed, each layer is gated against an independent source before the next is built on it.

Layer	Checked against	Result
Pool price	on-chain *slot0* state, exact integer tick math	reconstruction gate: pass
Fee accrual	the pool's own *feeGrowthInside* accumulator, read on-chain	relative error +0.18% / -0.52%
LVR	three independent constructions	sanity gate: pass
Price alignment	no observation dated after its own block	lookahead test: pass
No-arbitrage band	pool price against the reference exchange	median |gap| 3.9 bp on a 5 bp pool

Table 2. Validation gates. A failure stops the pipeline rather than producing a flagged number, on the principle that a figure computed on top of a broken stage is worse than no figure.

The no-arbitrage band deserves comment, because it is a result rather than a check. The median absolute gap between pool and reference is close to the fee tier on every pool — a few basis points on the 5 bp tier and of the order of the tier itself on the 30 bp tier. This is the fee band doing exactly what theory says it does: deviations smaller than the round-trip cost are not worth arbitraging, and the pool is left stale inside that band.

6. Results

6.1 Does providing liquidity pay?

Table 3 reports the decomposition, 43 monthly cohorts per pool except the gold pool's 16. The project's own stated falsification test was a pool in which fees persistently exceed LVR by more than costs. It is met, twice.

Pool	median σ	fees	LVR	gas	net	95% CI	months > 0
USDC/USDT 1bp	**2.3%**	**+10.3%**	**-1.6%**	**-0.27%**	**+8.43%**	**[+5.0%, +11.6%]**	**83%**
PAXG/USDC 5bp	19.5%	+17.2%	-18.5%	-0.46%	**-1.82%**	[-6.2%, +3.4%]	50%
WBTC/ETH 30bp	28.3%	+32.8%	-31.5%	-0.60%	**+0.71%**	[-1.6%, +3.4%]	44%
WBTC/USDC 30bp	40.0%	+61.1%	-56.3%	-0.76%	**+4.09%**	[-0.5%, +9.0%]	47%
ETH/USDC 5bp	52.3%	+88.8%	-96.8%	-1.17%	**-9.10%**	[-15.6%, -3.3%]	26%
ETH/USDC 30bp	52.4%	+96.7%	-97.9%	-1.14%	**-2.31%**	[-5.2%, +1.3%]	21%

Table 3. Decomposition of a $100k position at ±10%, annualised; 43 monthly cohorts per pool except PAXG/USDC, which has 16. Net is fees – LVR – gas and is already delta-hedged by the identity in Equation (1). Confidence intervals are block bootstrap over whole months. ETH/USDC 100bp is withheld throughout: a $100k position is 113% of its active liquidity, so its returns are not price-taking measurements.

The stablecoin pool earns a significantly positive net return at a size holding approximately 0.00% of active liquidity, so this is not a capacity artefact. The deepest ETH/USDC tier loses roughly nine per cent a year. Between those, the ordering is *not* monotone in volatility, and Section 6.4 is about why that matters more than it first appears.

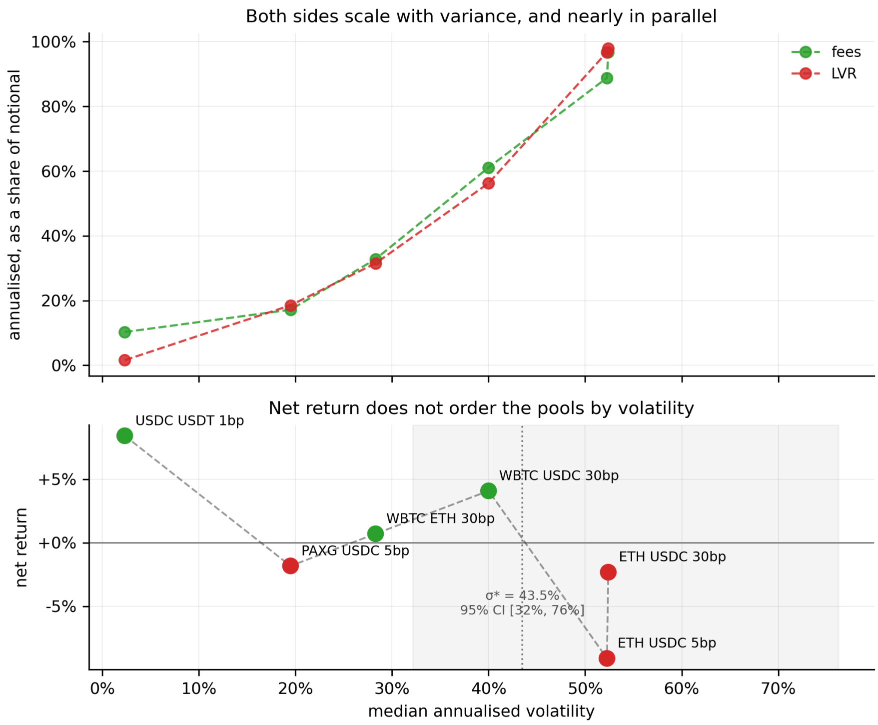

Figure 1. Net market-making return against median annualised volatility, seven pools. The negative slope across pools is real but, as Section 6.4 shows, is not a statement about what volatility does to a given pool.

6.2 Does the closed form hold?

We regress realised LVR on the closed form across 8,181 pool-days with Newey–West standard errors [6]. A slope of one and an intercept of zero would mean the theory is exactly right.

Pool	median σ	β	s.e.	R^2	p(β=1)	level match
USDC/USDT 1bp	2.3%	0.634	0.048	0.002	0.000	6.596
PAXG/USDC 5bp	**19.5%**	**0.718**	**0.182**	**0.208**	**0.121**	**1.064**
WBTC/ETH 30bp	28.3%	0.956	0.096	0.628	0.644	1.031
WBTC/USDC 30bp	40.0%	0.666	0.123	0.487	0.007	1.015
ETH/USDC 5bp	52.3%	0.784	0.089	0.618	0.015	0.972
ETH/USDC 30bp	52.4%	0.838	0.110	0.619	0.143	1.011

Table 4. Realised LVR regressed on the closed form, by pool. Level match is the ratio of summed realised to summed predicted LVR, which is insensitive to the slope and detects bias the regression can hide.

On the correlated-majors cross the null $\beta = 1$ cannot be rejected at all, the level matches within 3%, and R^2 is 0.628 — obtained with zero fitted parameters, on a pool the theory was not tuned to. On the stablecoin pool the relationship disappears entirely: R^2 is 0.002 and realised LVR is 6.6 times the prediction. That failure is the most informative result in this section, and it is not isolated.

6.3 The bias is monotone in volatility

Binning all pool-days by volatility rather than by pool shows that the closed form does not simply work or fail. It is biased in a single, monotone direction.

σ bucket	median σ	pool-days	realised / predicted
q1	**0.024**	**1,737**	**1.98**
q2	0.272	1,736	1.18
q3	0.433	1,736	1.08
p75-90	0.597	1,042	1.03
p90-99	0.843	625	0.96
top1%	**1.448**	**70**	**0.76**

Table 5. Ratio of realised to predicted LVR by volatility bucket, all seven pools pooled. The closed form under-predicts in calm markets and over-predicts in extreme ones.

Read as a curve rather than a list, this says something the per-pool table cannot. At low volatility the diffusive term nearly vanishes and exposes a **floor of adverse selection** the theory does not model — discrete blocks, jumps, and MEV-ordered flow all extract value that a diffusion with near-zero variance does not predict. At extreme volatility the closed form over-charges. The two failures previously reported separately, on the peg and on crash days, are the two ends of one curve, and the theory is unbiased between roughly 0.6 and 0.9 annualised.

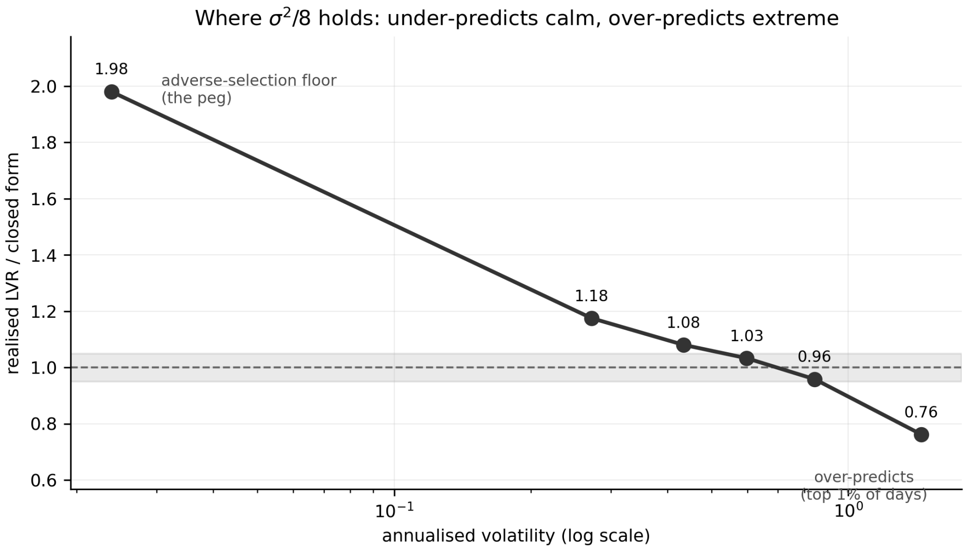

Figure 2. Realised over predicted LVR against volatility. The shaded band is ±5% around unbiased. Both known failures of the closed form are the ends of a single monotone curve.

This also dissolves a puzzle we carried for some time. The regression slope was 0.372 in 2023 against 0.918–0.944 in later years, which invited a regime explanation. It is not a year effect. The aggregate ratio for 2023 is 0.958, so the level was never wrong; only the slope was. By quarter the effect is confined to 2023Q3, whose five largest day-pool cells are 35.8% of the quarter's predicted LVR at a ratio of 0.182 — and all five fall on 17–18 August 2023. Removing the top percentile of volatility days moves 2023 to 0.801 and collapses the spread across years.

We tested three mechanisms for the extreme-day over-prediction and **all three were refused**. The no-arbitrage fee band predicts the ratio should rise with σ relative to the fee tier; it does not, and β is higher, not lower, at low σ/fee. Jumps predict that a jump-robust variance should fix it; substituting the bipower variation of Barndorff-Nielsen and Shephard [10] moves 2023 from 0.372 to 0.381. Microstructure oversampling predicts realised variance should inflate at fine sampling; it deflates.

The fourth candidate, exhaustion of the ±10% range during a large intraday move, is **refuted in the opposite direction**. The analytical rate already handles a range exactly, since |dn/dS| is taken on the position's own amount function and falls to zero at the boundaries, so any error must come from a mismatch of frequency: σ^2 is a daily aggregate while |dn/dS| is evaluated pointwise. That has a sharp test, because two days can share a variance and differ entirely in shape. Define

$$\textit{trend share} \;=\; |\Sigma\, r_t| \,/\, \sqrt{(\Sigma\, r_t^2)} \qquad (3)$$

which is 1 for a pure trend and near zero for a round trip. If converting out of the range were the mechanism, trending days would over-predict. They do the reverse.

Top 5% of volatility days	pool-days	median σ	realised / predicted
Price stays inside the range (oscillating)	175	1.120	**0.780**
Price leaves the range (trending)	28	1.186	**1.261**

Table 6. Realised over predicted LVR on the most volatile days, split by the shape of the move at comparable variance. Days on which the position converts out of its range are the ones the closed form gets *right*.

Regressing log(realised/predicted) on log σ and trend share over 4,044 pool-days gives a trend coefficient of +0.252 with a month-clustered 95% interval of [+0.212, +0.294] and P(coefficient < 0) = 0.000. The sign is unambiguous and the wrong one for range exhaustion.

The over-prediction is therefore concentrated on days whose variance comes from **oscillation rather than direction**. That points at discreteness on the extraction side rather than at the position: a pool can be arbitraged at most once per block, and movement smaller than the round-trip cost is never arbitraged at all, so a day that travels a long way without going anywhere generates realised variance that no arbitrageur can convert into profit. A directional move, by contrast, is captured in full. We report this as the surviving account rather than a demonstration: it is consistent with the sign, the magnitude and the tier evidence, but we have not measured extraction per block directly.

6.4 Volatility does not rank pools

The economic story usually told about LVR is that it scales with σ^2 while fee income does not, so volatile pools lose and calm pools win. Our own earlier drafts said exactly this, citing a cross-pool correlation between volatility and net return of –0.923. Adding a sixth pool broke it, and the tests that followed dismantled the mechanism.

The correlation is -0.569 across all seven pools and -0.730 across the six fitted. More decisively, WBTC/USDC earns +4.09% at 40.0% volatility while WBTC/ETH earns +0.71% at 28.3%. If volatility ranked pools, that could not happen.

Fitting net return on variance across 213 pool-months gives

$$net = a + b\cdot\sigma^2, \quad \sigma^* = \sqrt{(-a/b)} \tag{4}$$

with a crossover at σ* = 41.7% but a month-clustered 95% confidence interval of [30.4%, 77.3%] and R^2 of 0.048. There is no well-determined crossover to report, and the decile curve is not monotone. The reason is that both sides of the decomposition scale with variance:

$$fees = +0.043 + 3.06\cdot\sigma^2 \ \ (R^2 = 0.839), \quad LVR = -0.003 + 3.26\cdot\sigma^2 \ \ (R^2 = 0.921) \tag{5}$$

Volume rises with volatility, so fee income scales with variance almost as steeply as LVR does. Net return is the small difference between two large, near-parallel terms, which is precisely why its R^2 on variance is 0.055 and why the crossover interval spans forty points.

The cross-sectional association is also **Simpson's paradox** [11]. Estimating the same slope within pools, with pool fixed effects, gives +0.049 with a month-clustered interval of [-0.251, +0.379] and P(slope > 0) = 0.62. The sign flips and the estimate cannot be distinguished from zero; per-pool slopes are positive in four of five. The cross-pool pattern is therefore a statement about *which pools happen to be volatile*, not about what volatility does to a given pool, and it must not be read causally.

An earlier version of this paper offered a replacement: LVR per unit variance was 3.10–3.46 on every non-peg pool and 41.8 on the peg, so with the cost side that stable, *does this pool pay?* reduced to *is fee income per unit variance above roughly 3.3?* That ranked all five pools then in the sample perfectly. **The gold pool falsified it.**

Pool	median σ	LVR per unit variance
USDC/USDT 1bp	2.3%	**41.82**
PAXG/USDC 5bp	19.5%	**5.23**
WBTC/ETH 30bp	28.3%	3.10
WBTC/USDC 30bp	40.0%	3.32
ETH/USDC 5bp	52.3%	3.46
ETH/USDC 30bp	52.4%	3.34

Table 7. The cost side is not a constant off the peg. Ordered by volatility it is a continuum, and the pool added to test the claim landed between the two regimes the earlier reading had treated as the whole story.

Read in order of volatility this is not a constant with one exception but a single declining curve: 41.8 at 2.3%, **5.23 at 19.5%**, then 3.10–3.46 from 28% upward, where it flattens. The adverse-selection floor is present at every volatility and merely becomes small relative to the diffusive term once σ is high enough — the same shape as the bias curve of Section 6.3, measured on a different quantity.

The consequence for the ranking rule is direct. Spearman falls from +1.000 to +0.829, and the pool it misplaces is the one that broke the constant: gold clears a 3.3 threshold on fee income per unit variance (3.41) and still loses money, because its own cost side is 5.23. **A single threshold works only where LVR/σ^2 is flat, which is above roughly 28% annualised.** Below that the threshold is pool-specific and the rule collapses into the tautology that a pool pays when its fees exceed its costs. Two further caveats stand: the quantity is partly mechanical, since net is fees minus LVR; and lagged one month it does not forecast better than the naive volatility rule (76.2% of months positive against 78.6%).

6.5 A fee band does not reduce the cost

A natural conjecture is that a wider fee tier protects the provider, since it widens the no-arbitrage band inside which the pool is not worth attacking. Regressing LVR on variance separately by tier tests it directly.

Pool	fee tier	LVR per unit variance	s.e.	R^2
USDC/USDT 1bp	1 bp	0.00448	0.00033	0.002
PAXG/USDC 5bp	5 bp	0.00502	0.00124	0.211
ETH/USDC 5bp	5 bp	0.00553	0.00060	0.620
ETH/USDC 30bp	30 bp	0.00576	0.00073	0.624
WBTC/ETH 30bp	30 bp	0.00660	0.00059	0.630
WBTC/USDC 30bp	30 bp	0.00439	0.00086	0.471
ETH/USDC 100bp	100 bp	0.00590	0.00108	0.462

Table 8. LVR per unit variance by fee tier. Across a 100× range of fees the slope moves 1.50×, and the widest tier is not the lowest. The stablecoin pool is shown for completeness but its LVR is not variance-driven at all (R^2 = 0.002), so a slope is not a meaningful quantity there.

The premise does not survive. A wider band means each arbitrage is less frequent but larger, and to first order the two effects cancel. Monte-Carlo simulation of the reflected price gap reproduces the cancellation analytically. We report this as a negative result rather than a weak positive one.

6.6 Theory against machine learning

If the closed form captures the cost side, it should forecast LVR better than a flexible learner given the same information. We compare it against gradient boosting on the full feature set, out of sample under purged cross-validation.

Model	n out-of-sample	RMSE (USD)	OOS R^2	directional
$\sigma^2/8$ + HAR-RV	**352**	**1,347.9**	**+0.126**	**74.1%**
mean baseline	352	1,510.5	-0.097	50.0%
LightGBM	352	1,610.7	-0.247	56.5%
$\sigma^2/8$ + persistence	352	1,668.1	-0.338	72.4%

Table 9. Out-of-sample forecast of weekly LVR. A Diebold–Mariano test [7] of the theory against gradient boosting gives a statistic of –5.77 ($p < 0.0001$), favouring the closed form.

The closed form wins decisively, and the learner is worse than the sample mean. This is the cleanest positive result in the paper: a two-parameter-free expression from continuous-time theory beats a flexible model with access to the same data, because the quantity being forecast really is a function of variance.

6.7 Allocation, and where the edge is not

If the cost side is forecastable, an allocation across pools should be profitable. It is not. Table 8 reports weekly allocation over 165 weeks with full costs, deflated for the ten specifications logged [8].

Strategy	ann. return	volatility	Sharpe	max DD	deflated p
Always the stablecoin pool	+6.4%	3.2%	1.96	-3.4%	0.829
Always WBTC/USDC 30bp	+3.9%	4.4%	0.88	-5.7%	0.755
Our forecast (calibrated)	+1.5%	4.3%	0.36	-5.0%	0.198
Passive, equal weight	-0.3%	2.2%	-0.15	-3.3%	0.033
Always ETH/USDC 5bp	-10.1%	3.9%	-2.59	-27.4%	0.000

Table 10. Weekly allocation, five pools, 165 weeks, net of gas. Deflated probability is the Bailey–López de Prado statistic against ten logged trials; 0.95 is the bar. Perfect-foresight rows are excluded as a ceiling rather than a strategy: that ceiling is +17.3% at Sharpe 3.71.

Nothing clears the bar. An earlier version of this work recommended a static rule — always hold the lowest-volatility pool — at +6.4% and Sharpe 1.96. We now retire it, for three reasons that are worth stating because each is a trap the construction invites.

First, that headline was not implementable: it is *always hold the stablecoin pool*, a pool identified with hindsight, before gas. Selecting on lagged 30-day volatility and charging gas gives +4.31% at Sharpe 1.39.

Second, the Sharpe ratio is computed against zero rather than against cash, while the position ties up real capital. The realised 4-week Treasury bill, taken from the Treasury's own daily publication and carried forward from the last business day at or before each week end, averaged 4.62% over these weeks — more than the rule earns. Net of it the rule returns **-0.31% a year at a Sharpe of -0.10**, with a deflated probability of 0.039. It does not merely fail to clear the bar; it loses to Treasury bills.

Third, it is not a volatility rule. Walking down the volatility ladder, performance oscillates rather than decays: rank 1 +4.3%, rank 2 -1.2%, rank 3 +2.6%, rank 4 -6.2%. The second-lowest-volatility pool loses money while the first and third make it. A volatility mechanism would decay monotonically; this is pool selection, and it is the same conclusion Section 6.4 reached from the cross-section.

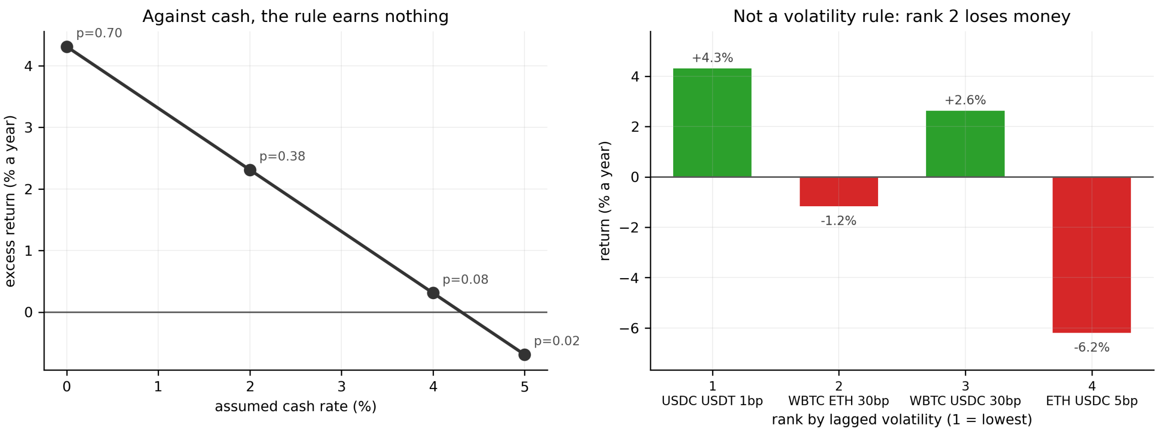

Figure 3. Left: the static rule's excess return against an assumed cash rate, with deflated probabilities annotated. Right: return by rank in lagged volatility. Rank 2 losing money while ranks 1 and 3 make it is the clearest single refutation of the volatility reading.

The negative result has a precise shape, and it is the paper's most useful practical finding. Replacing each component of the forecast with its own oracle shows that perfect variance forecasting is worth 0.9 percentage points a year, while perfect fee forecasting is worth 11.4 — about 80% of the gap to the ceiling. **The closed form prices the cost side well and says nothing about the income side, and the income side is where the uncertainty lives.** Allocating across liquidity pools is a volume-forecasting problem wearing a volatility costume, which is why a correct theory of the cost does not become a strategy.

Position specification is varied throughout and does not rescue this. Net return is monotonically increasing in range width — a ±1% range loses 70% a year against 3% for ±25% — because concentration multiplies fees and LVR by the same factor while multiplying gas by more. Concentration is sign-neutral on the market-making result and merely levers it.

6.8 What the MEV classification misses

MEV magnitude is the weakest quantity in this paper, and an earlier version said so on the grounds that bundle-level attribution needs relay data we could not obtain. That was wrong. The Flashbots relay does serve historical bidtraces; the earlier attempt queried them by consensus slot rather than by block number and read the empty responses as absence. What it publishes is the **payment from builder to proposer** for each delivered block — the price the builder was willing to pay to order that block, and therefore the market's own valuation of the extractable value in it.

Block group	queried	delivered	relay coverage	median payment
Contains a detected sandwich	**2,500**	**713**	**28.5%**	**0.3691 ETH**
Contains a swap, no sandwich	1,400	299	21.4%	0.0528 ETH

Table 11. Proposer payments for blocks in which the classifier does and does not find a sandwich. Coverage is well below 100% because Flashbots is one relay among several; a block delivered elsewhere is unobserved, not worthless, and is dropped rather than counted as zero. The control fetch was stopped early — the relay throttles progressively, from 0.2 seconds per block to roughly 35 — and the realised sample size is reported rather than the target.

Blocks in which we detect a sandwich carry **7.0 times** the proposer payment of blocks that merely contain a swap, with a bootstrap 95% interval of [5.7, 8.6]. That is an independent, market-set confirmation that the classifier is finding real extraction rather than pattern-matching noise, and it comes from a source with no connection to the on-chain heuristics that produced the labels.

On those same blocks the profit we attribute is **101%** of what the builder paid the proposer. We read this as economics rather than as a validation of magnitude. Our estimator is *gross*: it is the token value the attacker gained, with no deduction for the bid it had to place to obtain that position in the block. The proposer payment is essentially that bid. Gross profit of the same order as the bid is what a competitive searcher auction predicts — the rent is bid away to the proposer, and the searcher keeps a thin margin.

This sharpens the caveat rather than removing it, and it cuts in two directions. The **count** of sandwiches remains a lower bound: we observe one pool, from on-chain traces, on blocks delivered through one relay. But the **profit** attached to each detected sandwich should not be read as a transfer to searchers, because most of it is paid onward to the proposer. Neither quantity enters the decomposition, which is why the central results do not move; but the earlier framing of MEV as a purely one-sided undercount was too simple.

7. Three construction faults

Each fault below was found after results had been generated, none raised an error, none broke a test, and each materially changed a reported number. We document them because a silent specification error is the failure mode this measurement problem is most exposed to.

7.1 A stale cache gave an entire pool zero fees

The daily fee series is cached. The panel builder reused that cache whenever the file existed, and it had been built when the universe was five pools. Adding a sixth left it a universe behind: WBTC/USDC merged into the panel with **zero fees for all 188 weeks**, its net return became the negative of its LVR, and the allocation reported –57% a year with 0.0% of weeks positive. Nothing caught it, because every individual number was valid — zero fees is what a quiet day looks like, so no range check fires. It surfaced only because the monthly decomposition independently reported +4.09% for the same pool and both could not be true. The cache is now checked against the current pool universe rather than merely existing, and a pool reaching the panel with no non-zero fee row raises.

7.2 A signed integer was decoded at the wrong width

Uniswap v3 encodes ticks as *int24*, but the ABI sign-extends them to 256 bits. Decoding at the declared width returned tick –2 as a 78-digit positive integer. On ETH/USDC the fault was invisible, because that pool's ticks sit near +205,000 and never approach zero; on the stablecoin pool, whose ticks oscillate around zero, it was universal. It surfaced only because a columnar writer refused the oversized integer. Decoding now happens at 256 bits followed by an explicit range check against the declared width.

7.3 The hedge did not know what it was hedging

The hedging module shorted an ETH perpetual against whatever the position's risky delta returned. On a pool whose risky leg is not ETH that is a hedge against the wrong asset, and on a stablecoin pool it hedges a dollar exposure with a cryptocurrency. Funding costs reached −9,856% of notional, which is the only reason it was noticed. Pools now declare which legs carry dollar price risk, and the hedge refuses to run where that mapping is absent.

All three are covered by regression tests that fail if the specification returns. We note that the first was found by cross-checking two independent constructions of the same quantity, the second by a strict serialisation format, and the third by an implausible magnitude — three different mechanisms, none of them a unit test written in advance.

8. Limitations and falsification

Simulated positions. We follow a specified position rather than an observed provider. This is deliberate — pool aggregates cannot answer the question — but it means the results describe a policy, not a population of providers.

MEV is undercounted. Bundle-level attribution requires relay data no longer served for our period. The undercount is one-sided. Attribution is neutral with respect to the fees-minus-LVR identity, so the decomposition does not rest on it, but the MEV magnitudes should be read as a lower bound.

Six pools is a small cross-section. The volatility range is wide but the region between 2.3% and 28.3% annualised is unobserved entirely, and that gap is where the fee-income-per-unit-variance account would be tested hardest.

The extreme-day mechanism is identified but not measured. Four explanations are refused in Section 6.3 and the surviving account — discreteness on the extraction side — is inferred from the sign and magnitude of a shape effect rather than from per-block extraction, which we do not measure.

No risk-free series. The retirement of the static rule rests on a sensitivity band across plausible cash rates rather than a realised Treasury path. The qualitative conclusion is insensitive to the choice within that band, but the precise excess return is not pinned down.

We would regard the findings as falsified by: a pool whose LVR per unit variance sits far from 3.3 without being a peg; a demonstration that fee income per unit variance is predictable a month ahead, which would convert the negative result of Section 6.7 into a strategy; or a within-pool test on a longer sample that recovers a significant negative effect of volatility on net return.

9. Conclusion

Providing liquidity without an oracle has a price, that price has a closed form, and the closed form is right. Across seven pools spanning a 23× range of volatility it explains around 62% of daily variation in realised loss-versus-rebalancing with no fitted parameters, beats gradient boosting decisively out of sample, and has a bias that is monotone and therefore characterisable: it under-predicts in calm markets, where a floor of adverse selection it does not model dominates, and over-predicts in extreme ones.

Whether the price is worth paying depends on the pool. Liquidity provision is not uniformly loss-making — three of six pools have a positive point estimate and one is significantly positive — which falsified our own stated test. But the mechanism usually offered for this, that volatility determines the answer, does not survive. The cross-sectional association reverses sign within pools, and fee income scales with variance almost as steeply as the cost does. What separates pools is fee income per unit of variance, and volatility proxies for it only by coincidence of composition.

None of this becomes a strategy, and the reason is specific rather than a general appeal to efficiency. Decomposing the gap to a perfect-foresight ceiling shows that perfect variance forecasting is worth 0.9 percentage points and perfect fee forecasting 11.4. The theory addresses the side of the ledger that is already predictable. **The cost of providing liquidity is now measurable without assumption; the return to it is not forecastable, and the two facts have the same cause.**

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Data and code. The full pipeline reproduces every figure and table in this paper from raw swap events under a single command. All construction choices are configuration parameters with documented baselines and alternatives; every specification executed during the study is logged automatically; and the analysis is covered by 266 automated tests. Revision 02ebe78.